Finite-reference limits and evidence aggregation in sequential change detection

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Summary

When the nominal distribution is learned from a reference sample of size m, that sample caps the evidence available to a detector controlling the probability of ever raising a false alarm. We characterize sequential detection near this cap. In regular exponential families, the critical window is a central-limit neighbourhood of the testing boundary. Optimal power on the time scale $m^{3/2}$ has a Gaussian limit, and the delay has a reciprocal Gaussian limit with an atom at infinity. Calibrated generalized likelihood-ratio scans and proper mixtures over unknown change times attain these limits. For procedures that aggregate homogeneous-block mixture likelihoods, including specified e-detector constructions, we derive a sharp power envelope on the longer time scale m^2. Adjusted running-maximum procedures attain that envelope and the same eventual power as the unrestricted optimum. Their low detection quantiles differ by a factor of order $m^{1/2}$. Simulations exhibit this timing separation and substantial finite-sample corrections.

Some key words: Change detection; Composite null; Critical limit; Evidence aggregation; Finite reference; Sequential inference.

1. Introduction

An unknown nominal distribution must often be learned from a finite historical sample before surveillance begins. Increasing the monitoring horizon does not remove all uncertainty in that sample. Even after the changed distribution is known accurately, the historical observations may provide insufficient evidence to distinguish the changed experiment from a constant stream having that distribution. Under control of the probability of ever making a false alarm, this limitation determines both attainable power and the time needed to attain it.

We study independent observations whose distribution changes from a fixed P to a fixed Q after m observations. A detector must satisfy

$$\sup_{R\in\mathcal{P}} \mathrm{pr}_R(T<\infty)\le\alpha, \qquad b=\log(1/\alpha), \tag{1}$$

where under the null all observations have one common law R. The probability averages over the random reference sample. Let $d=\mathrm{KL}(P\|Q)$. Against the constant-Q null, the mean log likelihood ratio of the entire changed experiment is md. The critical regime is therefore $md-b$ of order $m^{1/2}$, where power need not approach either zero or one.

The information underlying the unrestricted optimum is

$$D(m,t)=\inf_{R\in\mathcal{P}}\{m\,\mathrm{KL}(P\|R)+t\,\mathrm{KL}(Q\|R)\}. \tag{2}$$

The corresponding homogeneous-block objective is the infimum of the maximum of the two terms. These objectives approach the same ceiling at different rates, and the difference persists at the

scale of the reference fluctuation. The information (2) has a familiar form: a weighted generalized Jensen–Shannon divergence for a full finite alphabet (Gutman, 1989; Haghifam et al., 2021), and Bregman information in an exponential family (Banerjee et al., 2005). At a fixed sample-size ratio, Zhou et al. (2020, Corollary 3) obtain the same second-order information and variance in two-sample homogeneity testing.

Our contributions concern the critical experiment and its dependence on evidence aggregation. Writing $md = b + c\varsigma\sqrt{m} + o(\sqrt{m})$, we show that optimal power by time $\lambda m^{3/2}$ converges to $\Phi\{c - \chi/(\lambda\varsigma)\}$, with a reciprocal Gaussian delay law carrying mass $1 - \Phi(c)$ at infinity. Two-segment mixtures, calibrated likelihood-ratio scans and proper mixtures over unknown splits attain this limit. For the homogeneous-block mixture class in Definition 1, we prove the sharp envelope $\Phi\{c - 2\sqrt{\chi d}/(\varsigma\sqrt{\lambda})\}$ on the scale m^2 and give attaining procedures. The optimal low detection quantiles of the two classes therefore differ by order $m^{1/2}$, despite their common eventual-power limit. Away from the critical window, Gaussian location gives the first-order block-to-two-segment delay ratio $(\sqrt{\kappa} + 1)/(\sqrt{\kappa} - 1)$, where $\kappa = md/b > 1$; it is approximately 5.8 at $\kappa = 2$.

Historical-sample monitoring with asymptotic false-alarm control was developed by Chu et al. (1996) and Horváth et al. (2004); detection-delay limits were studied by Aue & Horváth (2004); Aue et al. (2008). For fixed alternatives, the conventional regime with fixed α and $m \to \infty$ has $md/b \to \infty$. Here α decreases jointly with m, while the reference information remains within a central-limit scale of the boundary, and validity holds at each finite α. Finite Phase I estimation also affects Phase II control-chart performance (Jensen et al., 2006). Related surveillance analyses use other combinations of error criteria, reference asymptotics and signal assumptions (Pollak & Siegmund, 1991; Mei, 2006; Maillard, 2019; Yu et al., 2023; Kirch & Stoehr, 2022); Aue & Kirch (2024) review the broader role of type-I error control.

Likelihood domination and mixture constructions are established tools of sequential and universal inference (Wasserman et al., 2020; Hao et al., 2024; Kaufmann & Koolen, 2021). E-detectors and confidence-sequence reductions provide general monitoring frameworks (Shin et al., 2024; Shekhar & Ramdas, 2024). Closest to our setting, Saha & Ramdas (2026) establish an early-change relative-entropy bound and late-change first-order guarantees for non-partitioned e-detectors. Remark 1 identifies the mixture-likelihood instances covered by our block class, including a Bernoulli specialization of their construction. For these instances, our critical converse quantifies a delay penalty in a regime complementary to long-prefix asymptotics. Calibrated two-segment scans, including the Bernoulli scan of Besson et al. (2022), attain the unrestricted envelope. Recent conformal-martingale work also distinguishes finite-prefix obstructions from guarantees with sufficiently large reference information (Bhattacharyya & Ramdas, 2026).

All asymptotic power results are pointwise in a fixed distinct pair and fixed dimension; exact validity is uniform over the stated null family. Gaussian and Bernoulli experiments examine the power curves, evidence costs and timing separation. The observed limiting constants remain quantitatively offset in several finite cells. Selected-null calibration and paired grid and precision checks assess the implementations. The supplement gives complete proofs, model formulae and supporting empirical results.

2. EXPERIMENT AND INFORMATION

2.1. *Model and regularity*

For a declared reference size m, let $X_1, \ldots, X_m$ have law P and let the subsequent $Y_1, Y_2, \ldots$ have law Q, independently. A monitoring stopping time τ uses $\sigma(X_{1:m}, Y_{1:t})$ at monitoring time t; its absolute alarm time is $m + \tau$. Write $\mathcal{T}_\alpha(m)$ for rules satisfying (1), with independent auxiliary

randomization allowed. We use T for an absolute alarm time when the split is unknown. A missed alarm has time infinity.

The sharp constructive results concern a full minimal regular exponential family

$$f_\eta(z) = \exp\{\eta^\top T(z) - A(\eta)\}w(z), \qquad \eta \in \Theta \subset \mathbb{R}^p, \tag{3}$$

with common support and an open convex natural domain. Here $T(z)$ denotes the sufficient statistic; T without an argument denotes an alarm time. The dimension p is fixed. Put $\mu(\eta) = \nabla A(\eta)$, $V(\eta) = \nabla^2 A(\eta)$ and $F = A^*$. The distinct true laws $P = f_{\eta_P}$ and $Q = f_{\eta_Q}$ remain fixed as α decreases. Set $\mu_P = \mu(\eta_P)$, $\mu_Q = \mu(\eta_Q)$ and $\Delta = \mu_Q - \mu_P$.

Assumption 1. The mean domain contains a compact convex neighbourhood of the segment from μ_P to μ_Q, on which F is four times continuously differentiable with positive-definite Hessian. The corresponding natural parameters and natural segments used below lie in a compact subset of Θ, with a neighbourhood on which A is four times continuously differentiable and its Hessian is bounded above and below by positive-definite constant matrices. The fixed prior π is proper, with a continuous positive density on this natural neighbourhood. All suprema and integrals are measurable.

The neighbourhoods describe the fixed pair; they are not inputs to the procedure. A proper prior positive throughout Θ covers every pair satisfying the analytic assumptions. Interior natural parameters imply local exponential moments of the sufficient statistic. Infinite likelihood suprema are allowed, with the ratio of a finite marginal to an infinite supremum set to zero.

2.2. *Information geometry*

Define $a = \mathrm{KL}(Q\|P)$, $d = \mathrm{KL}(P\|Q)$ and

$$\varsigma^2 = (\eta_P - \eta_Q)^\top V(\eta_P)(\eta_P - \eta_Q), \qquad \chi = \tfrac{1}{2}\Delta^\top V(\eta_Q)^{-1}\Delta. \tag{4}$$

Both ς and χ are positive. The information in (2) can be written $D(m,t) = mh(t/m)$.

Lemma 1. *Under Assumption 1, the unique minimizing null in $D(m,t)$ has mean $\mu_s = (\mu_P + s\mu_Q)/(1+s)$, where $s = t/m$. Writing $R_s = f_{\eta_s}$ with $\eta_s = \nabla F(\mu_s)$,*

$$h(s) = F(\mu_P) + sF(\mu_Q) - (1+s)F(\mu_s), \tag{5}$$

$$h'(s) = \mathrm{KL}(Q\|R_s), \qquad h''(s) = -\frac{\Delta^\top \nabla^2 F(\mu_s)\Delta}{(1+s)^3}. \tag{6}$$

Consequently h increases strictly and is strictly concave, $h'(0) = a$, and

$$D(m,t) = md - \frac{\chi m^2}{m+t} + O\left(\frac{m^3}{(m+t)^2}\right), \qquad t/m \longrightarrow \infty. \tag{7}$$

The function D is positively homogeneous and concave in its two arguments.

For a full finite alphabet, $R_s = (P + sQ)/(1+s)$ and $D(m,t) = t\,\mathrm{GJS}(P,Q,m/t)$ under the convention of Zhou et al. (2020). In a general exponential family the projection matches the sufficient-statistic mean and need not equal a literal mixture distribution.

2.3. *A valid likelihood construction and its fluctuations*

For a finite sequence define

$$\mathfrak{q}_n(z_{1:n}) = \int \prod_{i=1}^n f_\eta(z_i)\pi(\eta)\,d\eta, \qquad \mathcal{L}^*(z_{1:n}) = \sup_\eta \prod_{i=1}^n f_\eta(z_i),$$

with both quantities one for an empty sequence. The declared-split statistic is

$$E_{m,t} = \frac{\mathfrak{q}_m(X_{1:m})\mathfrak{q}_t(Y_{1:t})}{\mathcal{L}^*(X_{1:m}, Y_{1:t})}, \qquad \tau^{\mathrm{mix}} = \inf\{t \geq 1 : E_{m,t} \geq e^b\}. \tag{8}$$

For each null R, its denominator is at least the likelihood under R. The numerator is the marginal of a proper two-segment path law, so its likelihood ratio to R is a nonnegative martingale from absolute time zero. Ville's inequality proves (1), including randomness in the reference. This is likelihood domination, as in universal inference (Wasserman et al., 2020); $E_{m,t}$ itself need not be a martingale.

Let $S_X = \sum_{i\leq m}\{T(X_i) - \mu_P\}$ and $S_Y = \sum_{j\leq t}\{T(Y_j) - \mu_Q\}$. Define

$$L_{m,t} = (\eta_P - \eta_s)^\top S_X + (\eta_Q - \eta_s)^\top S_Y, \tag{9}$$

$$V_{m,t} = m(\eta_P - \eta_s)^\top V_P(\eta_P - \eta_s) + t(\eta_Q - \eta_s)^\top V_Q(\eta_Q - \eta_s), \tag{10}$$

where $V_P = V(\eta_P)$ and $V_Q = V(\eta_Q)$. Write $\bar{T}_X, \bar{T}_Y, \bar{T}_Z$ for the two segment means and their pooled mean. The empirical likelihood gap is

$$G_{m,t} = mF(\bar{T}_X) + tF(\bar{T}_Y) - (m+t)F(\bar{T}_Z).$$

LEMMA 2. *For every deterministic $m, t \to \infty$, Assumption 1 gives*

$$G_{m,t} = D(m,t) + L_{m,t} + O_p(1), \qquad \log E_{m,t} = D(m,t) + L_{m,t} - \tfrac{p}{2}\log(mt) + O_p(1). \tag{11}$$

The logarithmic term is the classical Bayes-mixture penalty (Clarke & Barron, 1990); the supplement gives the uniform marginal bound used here. There are positive pair-dependent constants c_1, c_2 such that $c_1 D(m,t) \leq V_{m,t} \leq c_2 D(m,t)$, and $L_{m,t}/V_{m,t}^{1/2}$ converges in distribution to a standard normal random variable.

The likelihood ratio of the changed prefix to the deterministic null R_s has log exactly $D(m,t) + L_{m,t}$. For any valid rule and any real u, change of measure therefore gives

$$\mathrm{pr}_{P,Q}(\tau \leq t) \leq e^{u-b} + \mathrm{pr}_{P,Q}\{D(m,t) + L_{m,t} > u\}. \tag{12}$$

THEOREM 1. *Under Assumption 1, if $(D(m,t) - b)/V_{m,t}^{1/2} \to z \in \mathbb{R}$ and $\log(mt) = o(V_{m,t}^{1/2})$, then*

$$\sup_{\tau \in \mathcal{T}_\alpha(m)} \mathrm{pr}_{P,Q}(\tau \leq t) \longrightarrow \Phi(z), \qquad \mathrm{pr}_{P,Q}(\tau^{\mathrm{mix}} \leq t) \longrightarrow \Phi(z).$$

Here Φ is the standard normal distribution function.

The converse bounds a first-passage event, whereas attainment only needs a terminal likelihood exceedance. This distinction avoids treating a nonmonotone likelihood statistic as its running maximum. For a fixed finite-alphabet ratio, the information and variance agree with the second-order homogeneity result of Zhou et al. (2020); our use below includes a diverging ratio t/m.

3. THE UNRESTRICTED CRITICAL LIMIT

3.1. *The reference-information ceiling*

The eventual-power obstruction requires much less than an exponential family. Suppose $P \ll Q$, $Q \in \mathcal{P}$ and $d = \mathrm{KL}(P\|Q) < \infty$. Against the constant-$Q$ null the entire changed path has log likelihood ratio

$$R_m = \sum_{i=1}^{m} \log\{p(X_i)/q(X_i)\}.$$

Thus, for every real u and every globally valid rule,

$$\mathrm{pr}_{P,Q}(\tau < \infty) \le e^{u-b} + \mathrm{pr}_P(R_m > u). \tag{13}$$

If $\limsup md/b < 1$, the weak law gives eventual power tending to zero uniformly over valid rules. Finite variance is unnecessary for this conclusion. If $\mathrm{var}_P \log(p/q) = \varsigma^2 \in (0, \infty)$, the central limit theorem gives an upper envelope $\Phi(c)$ whenever

$$md = b + c\varsigma m^{1/2} + o(m^{1/2}), \qquad c \in \mathbb{R}. \tag{14}$$

The finite-prefix impossibility and a relative-entropy bound were established by Saha & Ramdas (2026, Theorem 7.3); (13) supplies the sharper subcritical conclusion and Gaussian envelope used here.

Under the regular model, the changed infinite path and the constant-Q path are equivalent for every finite m. A valid rule with $\alpha < 1$ therefore has positive probability of never alarming. Its unconditional expected detection time is infinite. We report power, finite quantiles and censored quantities, not an uncapped mean delay.

3.2. *Sharp power and delay*

Theorem 2. *Under Assumption 1 and* (14)*, for every fixed* $\lambda > 0$*,*

$$\sup_{\tau \in \mathcal{T}_\alpha(m)} \mathrm{pr}_{P,Q}\{\tau \le \lfloor \lambda m^{3/2} \rfloor\} \longrightarrow \Phi\left(c - \frac{\chi}{\lambda\varsigma}\right), \tag{15}$$

$$\sup_{\tau \in \mathcal{T}_\alpha(m)} \mathrm{pr}_{P,Q}(\tau < \infty) \longrightarrow \Phi(c). \tag{16}$$

The rule τ^{mix} *attains both limits. On* $[0, \infty]$*,*

$$\frac{\tau^{\mathrm{mix}}}{m^{3/2}} \Rightarrow W_D, \qquad W_D = \begin{cases} \chi/\{\varsigma(c+Z)\}, & c + Z > 0, \\ \infty, & c + Z \le 0, \end{cases} \quad Z \sim N(0,1). \tag{17}$$

Every valid rule has power tending to zero by $t = o(m^{3/2})$*. For* τ^{mix}*, power by any deterministic* t *with* $t/m^{3/2} \to \infty$ *tends to* $\Phi(c)$*.*

The last assertion concerns the stopping event, so it needs no restriction on $\log t$: monotonicity of that event and the eventual upper bound suffice. The limiting law is a reciprocal Gaussian expression with a mass at infinity, not an inverse Gaussian distribution.

To see the scale, (7) gives a deficit $\chi m^2/t$ from the ceiling. At $t = \lambda m^{3/2}$ it equals $(\chi/\lambda)m^{1/2}$, of the same order as the reference fluctuation. Moreover,

$$L_{m,t} = (\eta_P - \eta_Q)^\top S_X + O_p\{m^{3/2}/(m+t) + mt^{1/2}/(m+t)\},$$

and $V_{m,t}/m \to \varsigma^2$ as $t/m \to \infty$. Theorem 1 now gives (15). The eventual limit follows by letting λ increase and applying (13).

Away from the critical window, the same information describes first-order delay. If $\liminf md/b > 1$ and $\log m = o(b)$, define

$$t_D = mh^{-1}(b/m). \tag{18}$$

Then $\tau^{\mathrm{mix}}/t_D \to_p 1$, and every valid rule has power tending to zero by $(1-\epsilon)t_D$ for each fixed $\epsilon > 0$. For $m/b \to r > 1/d$, expansion of Theorem 1 also gives the ordinary delay central limit theorem. These first-order and Gaussian consequences are included in the supplement; the critical comparison below is the focus of the paper.

4. A CRITICAL PENALTY FOR HOMOGENEOUS-BLOCK AGGREGATION

4.1. *The restricted class and its information*

Now write the observations as one stream $Z_1, Z_2, \ldots$, with its first m observations from P. Let $\mathcal{F}_N = \sigma(Z_{1:N}, U)$ be its absolute-time filtration, where U contains any independent auxiliary randomization. For each start s, let

$$q_n^{(s)}(z_{s:s+n-1}) = \int \prod_{i=s}^{s+n-1} f_\xi(z_i)\pi_s(d\xi), \qquad M_{s:u}^\eta = \frac{q_{u-s+1}^{(s)}(Z_{s:u})}{\prod_{i=s}^u f_\eta(Z_i)}.$$

The proper prior π_s is independent of the evaluated null parameter η, is fixed before its block, and may depend on observations strictly before s. It cannot be selected retrospectively at the block endpoint. Its numerator fits one parameter throughout the block. Set $M_{s:s-1}^\eta = 1$ and $M_{s,N}^{\eta,*} = \max_{s-1\le u\le N} M_{s:u}^\eta$.

DEFINITION 1. *An $\mathcal{F}_N$-measurable nonnegative statistic S_N is block-dominated if, pathwise,*

$$S_N \le \inf_{\eta\in\Theta}\left\{c_N^\eta + \sum_{s=1}^N w_{s,N}a_{s,N}(M_{s,N}^{\eta,*})\right\}, \quad 0 \le c_N^\eta \le 1, \quad 0 \le w_{s,N} \le 1, \quad 0 \le a_{s,N}(x) \le x. \tag{19}$$

The weights, maps and c_N^η may depend on the data, provided the displayed quantities are measurable and their bounds hold pathwise. Its threshold rule is $T_S = \inf\{N : S_N \ge e^b\}$.

Validity is a separate requirement; the attaining procedures below are exactly valid. The definition permits one homogeneous-parameter mixture per start, with the pathwise aggregation bound (19).

Remark 1. With the block mixtures specified above, the following statistics are block-dominated:

$$\inf_\eta \sum_{s\le N} w_s M_{s:N}^\eta, \qquad \inf_\eta \sum_{s\le N} w_s a(M_{s,N}^{\eta,*}), \qquad \sum_{s\le N} \inf_\eta M_{s:N}^\eta, \qquad \max_{s\le N} \inf_\eta M_{s:N}^\eta,$$

where $0 \le w_s \le 1$ and $0 \le a(x) \le x$. For the last two, the sum of infima is at most the infimum of the sum, and a maximum is at most a sum. The first two instantiate the non-partitioned construction of Saha & Ramdas (2026, eqs. (1), (3) and Definition 6.3) with mixture likelihood-ratio blocks. A concrete covered example is their bounded-mean portfolio (18) on Bernoulli data: it becomes a beta$(1/2, 1/2)$ mixture likelihood ratio with a numerator independent of the candidate null. The last two use the SR and CUSUM aggregation of Shin et al. (2024, eq. (2.5)), instantiated with block e-processes $\inf_\eta M_{s:N}^\eta$ over the same null family. Theorem 3 applies at threshold e^b; its class bound requires no false-alarm guarantee. Their usual ARL calibration and probability-of-false-alarm calibration remain distinct. Membership here requires the stated mixture numerator; predictable plug-in products and mixtures whose numerators depend on the candidate null require separate analysis.

Let $K_P(\eta) = \mathrm{KL}(P\|f_\eta)$ and $K_Q(\eta) = \mathrm{KL}(Q\|f_\eta)$. Define

$$\mathcal{B}(m,t) = \min_\eta \max\{mK_P(\eta), tK_Q(\eta)\}. \tag{20}$$

The unique minimizer has mean $(1-z)\mu_P + z\mu_Q$, where $z \in (0,1)$ balances the two displayed terms. For every $m, t > 0$, $\mathcal{B}(m,t) < D(m,t)$. At that center, a contaminated full block cannot increase the maximum: the identity

$$mK_P(\eta) + tK_Q(\eta) = D(m,t) + (m+t)\,\mathrm{KL}(R_{t/m}\|f_\eta)$$

makes its value $2\mathcal{B} - D < \mathcal{B}$.

Lemma 3. *As $t/m \to \infty$, under Assumption 1,*

$$\mathcal{B}(m,t) = md - 2(\chi d)^{1/2}\frac{m^{3/2}}{t^{1/2}} + O(m^2/t). \tag{21}$$

At the balancing mean, $1 - z = (md/(\chi t))^{1/2} + O(m/t)$.

The deficit in (21) decays as $(m/t)^{1/2}$ relative to the ceiling, whereas the relative deficit of D decays as m/t. This suggests a quadratic critical horizon. Establishing the sharp power curve requires controlling fluctuations over every block; a first-order $o_p(b)$ bound does not suffice.

4.2. *A sharp block envelope and attainable laws*

Use one fixed prior satisfying Assumption 1, weights $w_s = 1/\{s(s+1)\}$ and the running-maximum adjuster

$$a(x) = \frac{2x}{(2 + \log x)^2}, \qquad x \geq 1. \tag{22}$$

The integral $\int_1^\infty a(x)x^{-2}\,dx$ equals one. Adjusters for running maxima originate in Shafer et al. (2011); Dawid et al. (2011); see Choe & Ramdas (2026) for their use across filtrations. Saha & Ramdas (2026, eq. (39)) use this same function for e-detectors. For inactive starts put $M^{\eta,*}_{s,N} = 1$. Define

$$S^{\mathrm{adj}}_N = \inf_\eta \sum_{s\geq 1} w_s a(M^{\eta,*}_{s,N}), \qquad T_{\mathrm{adj}} = \inf\{N : S^{\mathrm{adj}}_N \geq e^b\}, \tag{23}$$

$$S^0_N = \inf_\eta a(M^{\eta,*}_{1,N}), \qquad T_0 = \inf\{N : S^0_N \geq e^b\}. \tag{24}$$

The second rule retains only the running maximum from the beginning of the stream. Neither rule uses the true split. The first is the adjusted mixture-likelihood instance of Saha & Ramdas (2026, eq. (3), Definition 6.3), completed by the inactive contribution $1/\{2(N+1)\}$.

For each fixed null parameter, Ville's inequality and $\int_1^\infty a(x)x^{-2}\,dx = 1$ imply $\mathrm{E}_\eta\, a(M^{\eta,*}_{s,\infty}) \leq 1$. The terminal adjusted sum, or its origin-only counterpart, dominates the corresponding statistic at every time. Markov's inequality proves exact global validity of both rules. Their infimum processes need not be martingales.

Theorem 3. *Under Assumption 1 and* (14)*, let $t = \lfloor \lambda m^2 \rfloor$ for a fixed $\lambda > 0$. Every block-dominated statistic satisfies*

$$\max_{1\leq u\leq m+t} \log S_u \leq R_m - 2(\chi d/\lambda)^{1/2} m^{1/2} + o_p(m^{1/2}), \tag{25}$$

and consequently

$$\limsup \mathrm{pr}_{P,Q}(T_S \leq m + t) \leq \Phi\left(c - \frac{2(\chi d)^{1/2}}{\varsigma\lambda^{1/2}}\right). \tag{26}$$

For T_{adj} and T_0, the limit in (26) *exists and equals its right-hand side. Each has eventual power tending to $\Phi(c)$. For either attaining rule T_B, with delay $\Delta_B = (T_B - m)_+$,*

$$\frac{\Delta_B}{m^2} \Rightarrow W_B, \qquad W_B = \begin{cases} 4\chi d/\{\varsigma^2(c+Z)^2\}, & c + Z > 0, \\ \infty, & c + Z \leq 0, \end{cases} \quad Z \sim N(0,1). \tag{27}$$

Every block-dominated rule has power tending to zero by $m + o(m^2)$. Either attaining rule has power tending to $\Phi(c)$ by $m + t$ whenever $t/m^2 \to \infty$.

Proof. Let $\widehat{P}, \widehat{Q}$ be the exponential-family fits to the two empirical means, and let $\widehat{\eta}_B$ balance $m\,\mathrm{KL}(\widehat{P}\|f_\eta)$ and $t\,\mathrm{KL}(\widehat{Q}\|f_\eta)$. Their common value $\widehat{B}$ satisfies

$$\widehat{B} = R_m - 2(\chi d/\lambda)^{1/2} m^{1/2} + O_p(1). \tag{28}$$

The post-change sample moves the minimizing null by order m^{-1} beyond its deterministic order-$m^{-1/2}$ displacement from Q. Its effect on the minimized height is only $O_p(1)$; the leading fluctuation comes from the reference.

The key uniform bound is $\max_{s\le u\le m+t} \log M_{s:u}^{\widehat{\eta}_B} \le \widehat{B} + o_p(m^{1/2})$. For intervals with at most $m^{4/3}$ post-change observations, a conditional likelihood-ratio Markov bound controls the mixture relative to the true block law, and the reference score controls the result. For intermediate post-change counts, the fitted mean is near Q and the score stays a fixed fraction below md. For intervals with at least $t/2$ post-change observations, the gain from fitting a single parameter over using Q is $O_p(\log m)$, and the full post-change score controls every subinterval. Uniform concentration handles the random minimizing null. The logarithmic aggregation cost in (19) is negligible.

For the lower bound in (23), retain the two pure blocks. Their marginal, weight and adjuster costs are $O_p(\log m)$. For (24), retain the prefix ending at m and the full prefix ending at $m + t$. The full-prefix empirical mean differs from μ_Q by $O_p(m^{-1})$, so its empirical minimax value has the same expansion (28). The reference central limit theorem gives (26). Monotonicity of stopping events and the universal eventual bound give the remaining claims. All uniform estimates and the empirical-center argument are proved in the supplement. □

Thus attaining the same eventual-power ceiling does not imply attaining the same critical delay. With the normalizations

$$u = \frac{\varsigma t}{\chi m^{3/2}}, \qquad v = \frac{\varsigma^2 t}{4\chi d m^2}, \tag{29}$$

the two attained power curves are $\Phi(c - 1/u)$ and $\Phi(c - v^{-1/2})$. These curves use different physical time units.

COROLLARY 1. *Fix $0 < \rho < \Phi(c)$ and put $g_\rho = c - \Phi^{-1}(\rho) > 0$. Let $q_D(\rho)$ and $q_B(\rho)$ be the unconditional ρ-quantiles of the monitoring delays of τ^{mix} and either attaining block rule. Then*

$$q_D(\rho) \sim \frac{\chi}{\varsigma g_\rho} m^{3/2}, \qquad q_B(\rho) \sim \frac{4\chi d}{\varsigma^2 g_\rho^2} m^2, \tag{30}$$
$$\frac{q_B(\rho)}{q_D(\rho)} \sim \frac{4d}{\varsigma g_\rho} m^{1/2}.$$

The restriction $\rho < \Phi(c)$ is essential. Conditioning on detection would change the quantile level and should be reported explicitly.

4.3. *Relation to the ordinary delay regime*

When $m/b \to r > 0$ and $t/b \to v > 0$, the block class has the first-order envelope $\max_{u\le m+t} \log S_u \le \mathcal{B}(m,t) + o_p(b)$. The adjusted all-start rule (23) attains it. If $rd > 1$, its delay divided by the solution t_B of $\mathcal{B}(m, t_B) = b$ converges in probability to one. Theorem 3 establishes origin-only attainment in the critical regime; the finite-ratio attainment result here concerns T_{adj}.

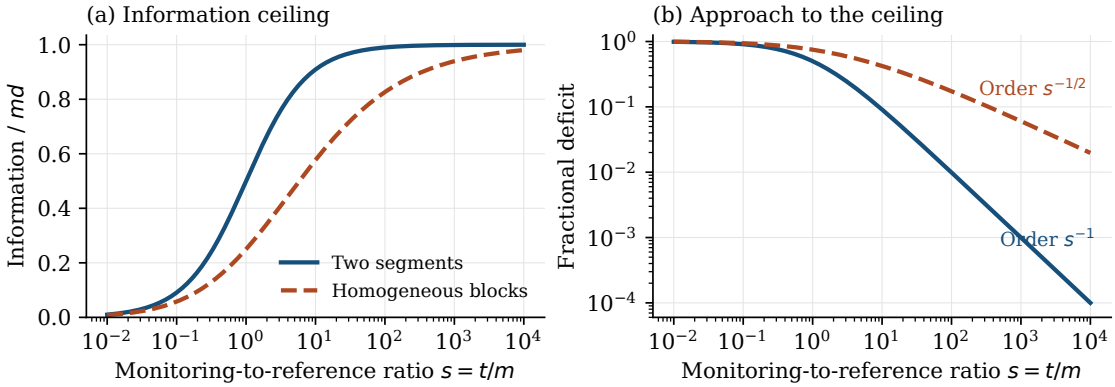


Fig. 1. Exact Gaussian information geometry. Left: attainable information as a fraction of the reference ceiling against $s = t/m$. Right: the deficits from that ceiling, with logarithmic axes. The two-segment deficit decays as s^{-1} and the block deficit as $s^{-1/2}$. Curves are analytic formulae, not empirical estimates.

Alt text: Two panels compare exact sum and maximum information. The sum lies above the maximum; its deficit decreases faster as the monitoring-to-reference ratio increases.

For Gaussian location with known covariance, put $I = \mathrm{KL}(P\|Q) = \mathrm{KL}(Q\|P)$. Here $\chi = I$, $\varsigma^2 = 2I$, and

$$D(m,t) = \frac{Imt}{m+t}, \qquad \mathcal{B}(m,t) = \frac{Imt}{(m^{1/2}+t^{1/2})^2}.$$

For $\kappa = mI/b > 1$ fixed,

$$\frac{t_D}{b/I} = \frac{\kappa}{\kappa-1}, \qquad \frac{t_B}{b/I} = \frac{\kappa}{(\kappa^{1/2}-1)^2}, \qquad \frac{t_B}{t_D} = \frac{\kappa^{1/2}+1}{\kappa^{1/2}-1}. \tag{31}$$

The different divergences as κ decreases to one anticipate the two critical exponents. Figure 1 shows the exact deterministic geometry; it contains no simulation results.

5. Existing likelihood tests and unknown splits

5.1. *A calibration bridge to generalized likelihood ratios*

The unrestricted optimum does not require the mixture numerator in (8). For a finite alphabet with K letters, let $C_K(n) = \binom{n+K-1}{K-1}$. The number of pairs of empirical distributions from samples of sizes m, t is $C_K(m)C_K(t)$. The method of types gives, for every common null distribution R, including boundary distributions,

$$\mathrm{pr}_R(G_{m,t} \geq x) \leq C_K(m)C_K(t)e^{-x}.$$

Consequently the declared-split rule with boundary

$$\beta_{m,t} = b + \log C_K(m) + \log C_K(t) + \log\{t(t+1)\} \tag{32}$$

has exact global false-alarm probability at most e^{-b}. At polynomial horizons this boundary exceeds b by $O(\log m)$, which is negligible on the critical scale.

For an unknown split at absolute time N, write $G_{j,N-j}$ for the two-sample likelihood gap and put

$$B_K(N) = \sum_{j=1}^{N-1} C_K(j)C_K(N-j), \qquad \beta_N = b + \log B_K(N) + \log\{N(N-1)\}. \tag{33}$$

Stopping when $\max_{1\le j<N} G_{j,N-j} \ge \beta_N$ also satisfies (1). For Bernoulli observations, $B_2(N) = (N-1)(N+1)(N+6)/6$.

PROPOSITION 1. *For distinct fixed interior finite-alphabet laws, both* (32) *and* (33) *attain the critical power and delay limits of Theorem 2. For the scan use absolute alarm time* $m + \lfloor \lambda m^{3/2} \rfloor$. *The theoretically calibrated Bernoulli generalized likelihood-ratio scan of Besson et al. (2022, Lemma 2) attains the same limits.*

At a polynomial critical horizon all these boundaries equal $b + O(\log m)$. The scan contains the true split, so Lemma 2 supplies a terminal lower bound and the universal converse supplies the upper bound. The supplement gives a self-contained type-counting proof, deterministic-calendar variants, a scalar known-variance Gaussian calibration, and the precise threshold comparison with Besson et al. (2022). Two-segment scans therefore lie on the unrestricted side of the separation in Theorem 3.

5.2. *A proper mixture over unknown splits*

For each integer $j \ge 1$, define a complete path law with marginals

$$q_{j,N}(z_{1:N}) = \begin{cases} \mathfrak{q}_N(z_{1:N}), & N \le j, \\ \mathfrak{q}_j(z_{1:j})\mathfrak{q}_{N-j}(z_{j+1:N}), & N > j. \end{cases}$$

It draws an independent parameter from the same prior at its predetermined split. With deterministic positive weights ω_j summing to one, let

$$E_N^{\mathrm{cp}} = \frac{\sum_{j\ge1} \omega_j q_{j,N}(Z_{1:N})}{\mathcal{L}^*(Z_{1:N})}, \qquad T_{\mathrm{cp}} = \inf\{N : E_N^{\mathrm{cp}} \ge e^b\}. \tag{34}$$

For $\omega_j = 1/\{j(j+1)\}$ the inactive numerator contribution is $\mathfrak{q}_N(Z_{1:N})/N$. It must not be replaced by constant capital one.

THEOREM 4. *The rule* (34) *satisfies* (1). *If the true pre-change length is m, then*

$$\log E_{m+t}^{\mathrm{cp}} \ge \log E_{m,t} - \log(1/\omega_m).$$

Under Assumption 1 and (14), *if* $\log(1/\omega_m) = o(m^{1/2})$, *the delay* $(T_{\mathrm{cp}} - m)_+$ *has the same critical power, eventual power and extended delay limit as* τ^{mix}. *Polynomial weights meet this condition. Under the same assumption, analogous first-order attainment holds when* $\liminf md/b > 1$, $\log m = o(b)$ *and* $\log(1/\omega_m) = o(b)$.

Validity follows from a mixture of complete path laws, including their pre-split marginals. The true-split component is a proof device; evaluating it alone does not implement the unknown-split procedure. The statements concern specified sequences of change times, not a uniform absence of a timing penalty for arbitrarily late changes.

5.3. *Deterministic thinning*

Suppose a retained split is $j = m - r \le m$, with $r = o(m), o(t)$, and let $E_{m+t}^{(j)}$ use that split in its numerator. The common denominator cancels. At population means the exact information lost

is $D(r,t)$, and

$$\log E_{m,t} - \log E^{(j)}_{m+t} = D(r,t) + O_p(r^{1/2}+1) = rd + O\{r^2/(r+t)\} + O_p(r^{1/2}+1). \quad (35)$$

Thus $r = o(m^{1/2})$ and an inverse-weight log penalty $o(m^{1/2})$ preserve the critical limit. The same split condition preserves the finite-ratio second-order envelope; $r = o(b)$ suffices for first-order limits when $m \asymp b$.

For example, choose $\epsilon_b = \min(1, b^{-3/4})$, let $j_1 = 1$, $j_{i+1} = \lceil (1+\epsilon_b) j_i \rceil$, and assign weights $1/\{i(i+1)\}$ to these retained indices. When $m \asymp b$, the preceding retained split has gap $O(b^{1/4})$ and log weight cost $O(\log b)$. There are $O\{1 + \epsilon_b^{-1} \log(eN)\}$ retained splits up to N. This counts stored sufficient statistics and marginal evaluations per checkpoint; it is an arithmetic-time bound only when one marginal evaluation has bounded cost. Sparse monitoring itself has substantial precedent (Besson et al., 2022; Moen, 2026); (35) identifies the accuracy required for these sharp limits.

The origin-only block rule needs no split dictionary. A deterministic checkpoint calendar preserves Theorem 3 if it retains a time at or before m with gap $o(m^{1/2})$ and, for every fixed $\lambda > 0$, a time at or before $m + \lambda m^2$ with gap $o(m^2)$. For example, $N_i = \lfloor i^{3/2} \rfloor$ meets both conditions without knowing m. For the two-segment critical curve, monitoring gaps must be $o(m^{3/2})$. Exact validity survives all these deterministic restrictions. The supplement proves the preservation claims; numerical minimization must still use a conservative certificate for an infimum.

6. Empirical assessment

6.1. *Design and reporting*

We examine the critical laws, their finite-sample costs, and the ordinary finite-ratio comparison in five simulation studies. C1 uses Gaussian location and Bernoulli observations with a declared split: there are 5000 independent paths per Gaussian cell and 1000 per Bernoulli cell, across 14 cells with $b = 160, 640, 2560$. C2 compares unknown-start procedures for Bernoulli probabilities $0.2 \to 0.5$, with 1000 paired paths in each of five cells at $b = 40, 160, 640$. C3 uses Bernoulli probabilities $0.3 \to 0.5$, $b = 20, 80$ and $\kappa = md/b$ near 2 or 4, with 500 paired paths per cell. C4 checks selected nulls at moderate error levels, and C5 examines grids and numerical precision on two additional 100-path cohorts. Together these studies contain 65 200 independent paths. Methods evaluated on the same path are paired, not additional replications. The Supplementary Material gives the completed design, further figures, numerical summaries and implementation details.

For critical cells we use the realized coordinate $c_m = (md - b)/(\varsigma\sqrt{m})$ and the scales $s_D = (\chi/\varsigma)m^{3/2}$ and $s_B = (4\chi d/\varsigma^2)m^2$. Thus the limits in (29) are evaluated at $u = t/s_D$ and $v = t/s_B$. The large boundaries investigate the asymptotic experiment; null calibration is assessed separately at $\alpha = 0.1, 0.01$. Empirical first-alarm distribution functions retain every path in their denominator. Their intervals are pointwise 95% Wilson intervals; quantile intervals use 2000 paired bootstrap resamples. Unreached quantiles and unidentified interval endpoints remain censored. All curves use actual inspection checkpoints, and terminal proportions estimate detection by the finite cap, not eventual power. No exponent is fitted to the few available boundary sizes.

6.2. *Critical power and finite-sample costs*

Figure 2 shows the central C1 cells. The Gaussian mixture and calibrated GLR-G have similar power and move toward the limiting curve as the boundary increases. Bernoulli GLR-B and GLR-T have larger, distinct finite-threshold penalties. Here GLR-B uses the theoretical calibration of

Besson et al. (2022); GLR-T uses type counting, and GLR-G uses the scalar Gaussian calibration. The alternative oracle has privileged parameters and serves only as an analysis reference.

The raw limiting curves are not yet quantitatively accurate in every cell. At $b = 2560$, $c_m \simeq 0$ and the checkpoint near $u = 8$, Gaussian mixture power is 0.3756, compared with 0.4507 for the asymptotic target; GLR-G power is 0.3820. The corresponding Bernoulli mixture, GLR-B and GLR-T proportions are 0.393, 0.079 and 0.257. These differences do not contradict the shared asymptotic attainment in Proposition 1.

Saved-path decompositions separate marginal-likelihood regret and boundary costs. An exploratory Gaussian shift using the mean normalized regret on ten preselected nested paths gives 0.3737 for the Gaussian mixture at the above checkpoint. Such shifts explain much of the displacement, but they share data with the empirical analysis, omit other finite-sample effects, and are not independently validated power approximations. The Supplementary Material retains the unshifted targets, full decompositions and diagnostic uncertainty conventions.

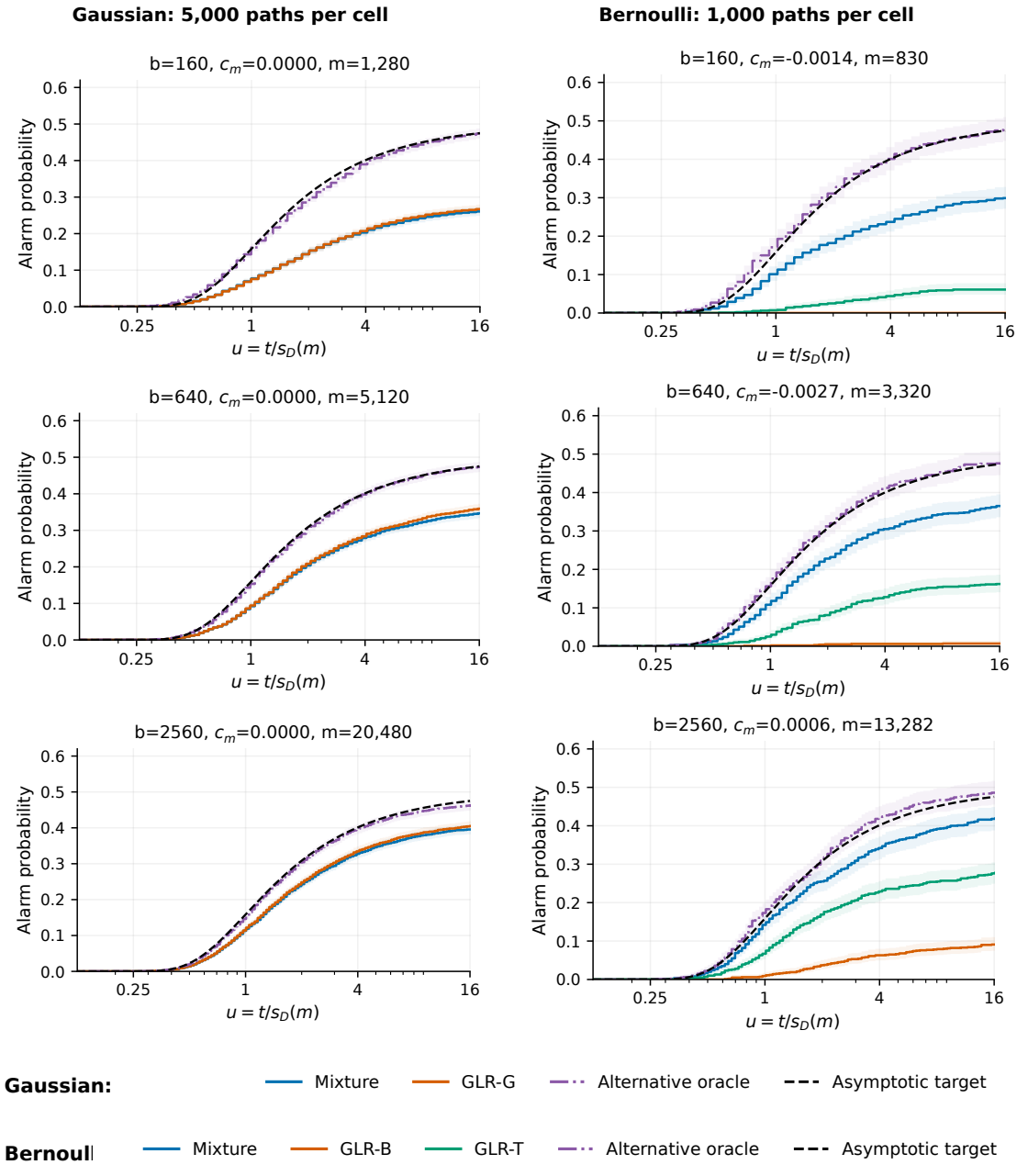


Fig. 2. Declared-split critical power at $c_m \simeq 0$: Gaussian location (left) and Bernoulli observations (right), with 5000 and 1000 independent paths per cell, respectively. Rows use $b = 160, 640, 2560$. Curves are all-path first-alarm proportions at actual checkpoints; shading gives pointwise 95% Wilson intervals. The asymptotic target is $\Phi(c_m - 1/u)$. The oracle uses the alternative parameters.

Alt text: Across increasing boundaries, the Gaussian mixture and GLR curves approach the limiting critical curve. Bernoulli calibrations have larger remaining offsets.

6.3. *The critical evidence-aggregation penalty*

In C2 the complete unknown-start mixture is compared with the origin-only adjusted rule and both Bernoulli GLRs on the same paths and calendar. Figure 3 displays the different timing scales. At $b = 640$, $c_m \simeq 0$, mixture power near $u = 8$ is 0.200 versus the limiting value 0.4494; origin power near $v = 8$ is 0.180 versus 0.3610. These checkpoints have different physical times. The largest cell has $m = 3320$ and an actual post-change cap of 39 936 973 observations.

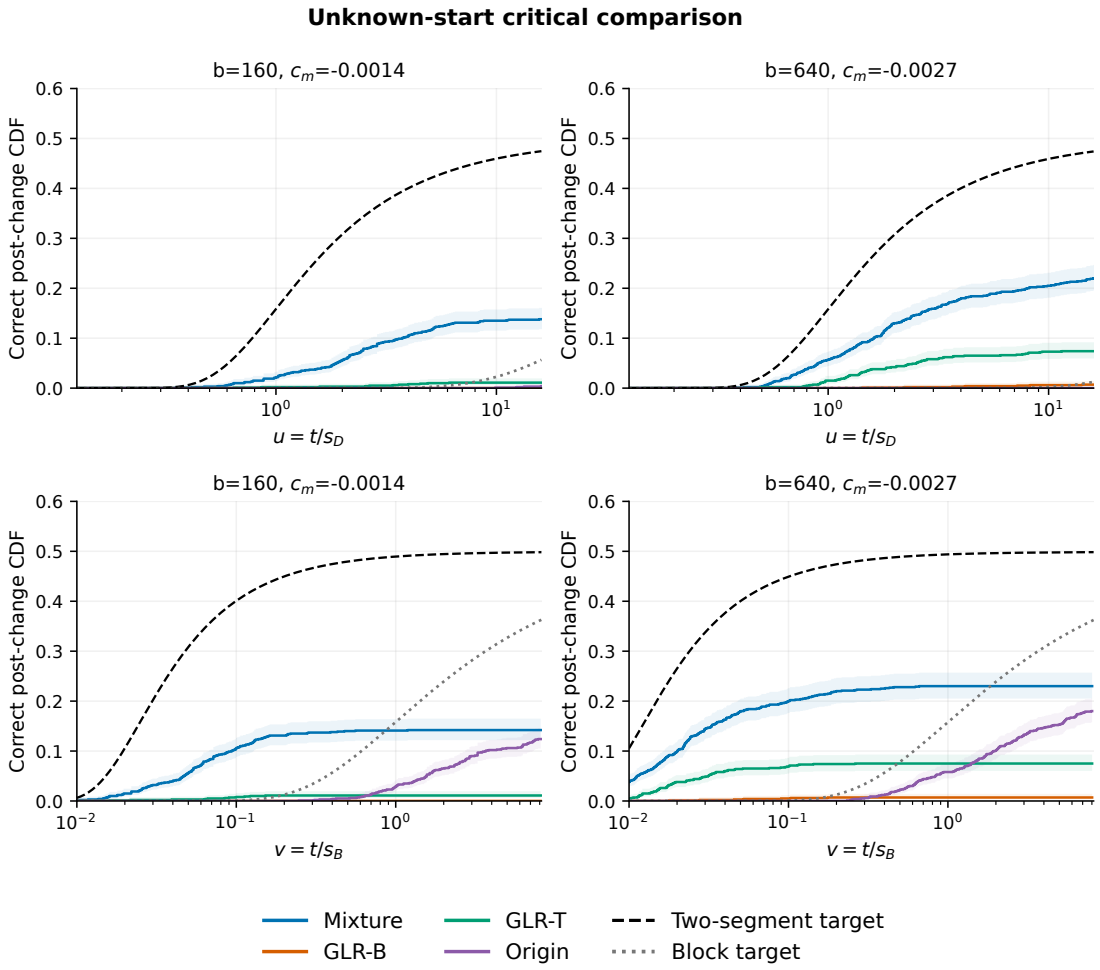


Fig. 3. Unknown-start Bernoulli critical comparison at $c_m \simeq 0$ and $b = 160, 640$, with 1000 paired paths per cell. The same first-alarm outcomes are shown on the two-segment scale $u = t/s_D$ (top) and block scale $v = t/s_B$ (bottom). Both targets are converted to the displayed coordinates. Shading gives pointwise 95% Wilson intervals. All paths remain in the denominator; observed pre-change alarms are zero. The full three-boundary comparison is in the Supplementary Material.

Alt text: The mixture begins detecting earlier than the origin rule, while both retain substantial downward offsets from their limiting curves at the simulated boundaries.

The unconditional 5% origin-to-mixture quantile ratios are 6.93, 33.72 and 74.80 as b increases (Fig. 4). The smallest-cell bootstrap envelope is unidentified. The two larger envelopes are $[28.74, 39.16]$ and $[59.54, 94.28]$, whereas the corresponding asymptotic ratios from Corollary 1 are 24.37 and 48.79. Between $b = 160$ and 640, the ratio grows by a factor of 2.22, compared with 2 from the leading $m^{1/2}$ scaling. Its level is 38–53% above the asymptotic benchmark, and its normalized value increases from 0.84 to 0.93, against a limiting value near 0.61. These two identified cells demonstrate timing separation but cannot distinguish a slowly converging constant from a different exponent; the exponent follows from Theorem 3. All central-cell 25% quantiles exceed the cap. At $b = 40$, the origin terminal proportion is 0.080 and the mixture proportion is 0.056. The procedures incur different marginal-likelihood and adjustment costs, and only the complete mixture pays split weights; their separate effects on this reversal are not identified by the comparison.

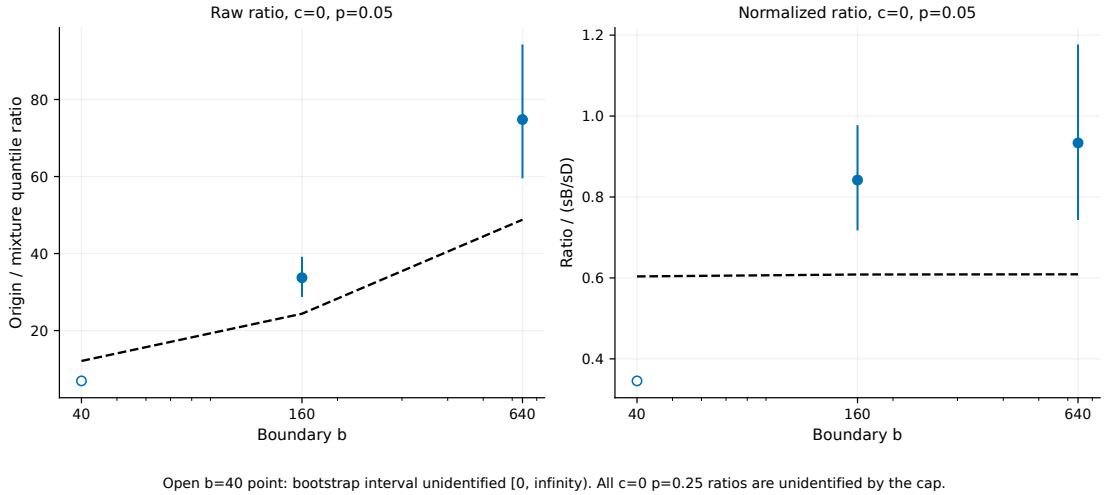


Fig. 4. Unconditional 5% origin-to-mixture quantile ratios at $c_m \simeq 0$. The right panel divides the ratio by s_B/s_D. Dashed curves are the realized-coordinate benchmarks from Corollary 1. Intervals use the original 2000 paired bootstrap resamples. At $b = 40$, 371 resamples are unidentified and the interval is $[0, \infty)$, despite a finite point estimate. The larger-cell intervals lie above the asymptotic constants.

Alt text: The observed low-quantile ratio grows from about seven to seventy-five; its normalized value increases between the two larger sizes and remains above the limiting benchmark.

The exact unknown-start mixture decomposition retains split weights, the signed difference between the true-split and selected-component likelihood statistics, and the full-sum aggregation gain. The origin diagnostic separates marginalization from the running-maximum adjustment. At the largest boundary, their exploratory shifted power values are 0.1961 and 0.1935, respectively. These diagnostics use ten nested paths per cell and do not establish convergence of the quantile constants. The class-wide converse is supplied by Theorem 3; the simulations compare its specified attainer with particular unrestricted procedures.

6.4. *The ordinary finite-ratio regime*

C3 uses the complete canonical adjusted restart comparator to assess the $\mathcal{B}$ limit of Section 4.3. The origin-only rule is not substituted for it. At $b = 80$, the canonical median delay is 6.741

or 3.323 times the mixture median at requested $\kappa = 2$ or 4 (Table 1). Both reported bootstrap envelopes exclude the corresponding population-root ratios. Normalized medians move toward one from $b = 20$ to 80, but finite-threshold offsets remain. The roots are calculated from the exact Bernoulli information functions, not the Gaussian formula (31).

Table 1. *Ordinary-regime median delays at $b = 80$, based on 500 paired Bernoulli paths per cell. Ratios are accompanied by 95% paired-bootstrap envelopes. Population-root ratios are first-order benchmarks.*

Requested κ	Mixture	Canonical	Ratio	Interval	t_B/t_D
2	3012	20 304	6.741	[6.386, 7.039]	5.862
4	1691	5620	3.323	[3.119, 3.436]	3.022

6.5. *Calibration, computation and practical error levels*

C4 checks Bernoulli and Gaussian nulls at $\alpha = 0.1, 0.01$; all 24 pointwise Wilson upper limits lie below the nominal levels. C5 gives identical decisions in all 900 stricter-certificate comparisons, while grid refinement changes distribution functions by up to three and four percentage points at the critical and ordinary anchors. The supplement gives the denominators, intervals and computational costs.

The power studies use $b \geq 20$, corresponding to $\alpha \leq e^{-20}$, to investigate the asymptotic regimes. Power at conventional error levels such as $\alpha = 0.01$ remains unassessed by these studies. At such levels the marginal, split-weight and adjustment costs can be comparable to b; the first-order delay ratios therefore provide benchmarks rather than finite-sample performance guarantees. C4 assesses null calibration at moderate levels, separately from power under a change.

7. DISCUSSION

Finite-reference monitoring has two distinct limitations. Reverse divergence controls how much evidence the entire experiment can contain, while evidence aggregation controls how quickly a procedure approaches that ceiling. In the critical window, homogeneous-block aggregation incurs a diverging delay penalty. An adjusted block procedure nevertheless attains the same eventual power as an unrestricted procedure. Origin-only attainment shows that the critical limitation lies in the form of evidence aggregation and persists without restarting.

Calibrated two-segment scans attain the unrestricted envelope. The operative distinction is therefore between two-segment and homogeneous-block evidence. Remark 1 connects the restricted class to existing mixture-based e-detectors. Combining complementary block evidence multiplicatively, or placing a change model inside a block numerator, leaves that class.

The experiments exhibit substantial timing separation and material finite-sample costs. In C2 the two larger normalized quantile ratios increase away from the asymptotic benchmark, even though their growth is close to the predicted square-root scaling. Marginalization, split weights and adjustment remain substantial relative to the reference fluctuation. The cost decompositions describe much of the power displacement; improved finite-sample approximations and procedures with smaller evidence costs are useful next steps. The current power studies address small nominal error probabilities, so conventional-level power comparisons remain a practical extension.

Extensions to shrinking changes, growing dimension, dependent streams and adaptive sampling require control of likelihood regret and concentration on the relevant critical scale. Adaptive allocation of reference and monitoring observations is another natural direction. The present

empirical checks concern fixed priors and two sensitivity anchors; broader implementation studies would clarify their operating range.

Supplementary material

The supplement gives complete proofs, the uniform block argument, exact scan calibrations, deterministic-grid results and closed-form model calculations. It also contains the full C1–C5 design, supporting critical and finite-ratio results, evidence-cost diagnostics, calibration tables and numerical-sensitivity analyses.

Declaration of the use of generative AI and AI-assisted technologies

Author review required before submission

ChatGPT (OpenAI) was used for literature comparison, to propose and check mathematical derivations, to audit arguments, and to draft, edit and typeset the manuscript. The authors must complete their verification of every proof, numerical result and reference, list any additional tools used, and finalize the journal's responsibility statement. The cover letter should contain a consistent disclosure.

Acknowledgement

Author information pending

Insert acknowledgements and funding information, or state that there are none, after author confirmation.

Data availability

Numerical summary tables and scripts for preparing the figures accompany the manuscript.

Repository information pending

Before submission, provide the permanent repository identifier for the simulation code, frozen configurations, random seeds and pathwise outputs. Those production materials were not supplied with this revision; the accompanying summaries and figure assets do not by themselves reproduce the original simulations.

References

Aue, A. & Horváth, L. (2004). Delay time in sequential detection of change. *Statistics and Probability Letters* **67**, 221–231.

Aue, A., Horváth, L., Kokoszka, P. & Steinebach, J. (2008). Monitoring shifts in mean: asymptotic normality of stopping times. *TEST* **17**, 515–530.

Aue, A. & Kirch, C. (2024). The state of cumulative sum sequential changepoint testing 70 years after Page. *Biometrika* **111**, 367–391.

Banerjee, A., Merugu, S., Dhillon, I. S. & Ghosh, J. (2005). Clustering with Bregman divergences. *Journal of Machine Learning Research* **6**, 1705–1749.

Besson, L., Kaufmann, E., Maillard, O.-A. & Seznec, J. (2022). Efficient change-point detection for tackling piecewise-stationary bandits. *Journal of Machine Learning Research* **23**, 1–40.

Bhattacharyya, S. & Ramdas, A. (2026). Change detection with conformal martingales: new optimal constructions, and suboptimality of existing methods. ArXiv:2609.27179v1.

Choe, Y. J. & Ramdas, A. (2026). Combining evidence across filtrations. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **88**, 1492–1514.

CHU, C.-S. J., STINCHCOMBE, M. & WHITE, H. (1996). Monitoring structural change. *Econometrica* **64**, 1045–1065.

CLARKE, B. S. & BARRON, A. R. (1990). Information-theoretic asymptotics of Bayes methods. *IEEE Transactions on Information Theory* **36**, 453–471.

DAWID, A. P., DE ROOIJ, S., SHAFER, G., SHEN, A., VERESHCHAGIN, N. & VOVK, V. (2011). Insuring against loss of evidence in game-theoretic probability. *Statistics and Probability Letters* **81**, 157–162.

GUTMAN, M. (1989). Asymptotically optimal classification for multiple tests with empirically observed statistics. *IEEE Transactions on Information Theory* **35**, 401–408.

HAGHIFAM, M., TAN, V. Y. F. & KHISTI, A. (2021). Sequential classification with empirically observed statistics. *IEEE Transactions on Information Theory* **67**, 3095–3113.

HAO, Y., GRÜNWALD, P., LARDY, T., LONG, L. & ADAMS, R. (2024). E-values for k-sample tests with exponential families. *Sankhya A* **86**, 596–636.

HORVÁTH, L., HUŠKOVÁ, M., KOKOSZKA, P. & STEINEBACH, J. (2004). Monitoring changes in linear models. *Journal of Statistical Planning and Inference* **126**, 225–251.

JENSEN, W. A., JONES-FARMER, L. A., CHAMP, C. W. & WOODALL, W. H. (2006). Effects of parameter estimation on control chart properties: a literature review. *Journal of Quality Technology* **38**, 349–364.

KAUFMANN, E. & KOOLEN, W. M. (2021). Mixture martingales revisited with applications to sequential tests and confidence intervals. *Journal of Machine Learning Research* **22**, 1–44.

KIRCH, C. & STOEHR, C. (2022). Asymptotic delay times of sequential tests based on U-statistics for early and late change points. *Journal of Statistical Planning and Inference* **221**, 114–135.

MAILLARD, O.-A. (2019). Sequential change-point detection: Laplace concentration of scan statistics and non-asymptotic delay bounds. In *Proceedings of the 30th International Conference on Algorithmic Learning Theory*, vol. 98 of *Proceedings of Machine Learning Research*.

MEI, Y. (2006). Sequential change-point detection when unknown parameters are present in the pre-change distribution. *The Annals of Statistics* **34**, 92–122.

MOEN, P. A. J. (2026). A grid-based methodology for fast online changepoint detection. ArXiv:2504.09573v2.

POLLAK, M. & SIEGMUND, D. (1991). Sequential detection of a change in a normal mean when the initial value is unknown. *The Annals of Statistics* **19**, 394–416.

SAHA, A. & RAMDAS, A. (2026). Non-partitioned e-detectors for nonparametric sequential change detection. ArXiv:2607.28322v1.

SHAFER, G., SHEN, A., VERESHCHAGIN, N. & VOVK, V. (2011). Test martingales, Bayes factors and p-values. *Statistical Science* **26**, 84–101.

SHEKHAR, S. & RAMDAS, A. (2024). Reducing sequential change detection to sequential estimation. In *Proceedings of the 41st International Conference on Machine Learning*, vol. 235 of *Proceedings of Machine Learning Research*.

SHIN, J., RAMDAS, A. & RINALDO, A. (2024). E-detectors: a nonparametric framework for sequential change detection. *New England Journal of Statistics in Data Science* **2**, 229–260.

WASSERMAN, L., RAMDAS, A. & BALAKRISHNAN, S. (2020). Universal inference. *Proceedings of the National Academy of Sciences* **117**, 16880–16890.

YU, Y., MADRID PADILLA, O. H., WANG, D. & RINALDO, A. (2023). A note on online change point detection. *Sequential Analysis* **42**, 438–471.

ZHOU, L., TAN, V. Y. F. & MOTANI, M. (2020). Second-order asymptotically optimal statistical classification. *Information and Inference: A Journal of the IMA* **9**, 81–111.